

Predicting Neighborhood Transit Recovery: Analyzing NYC Subway Ridership Patterns and Economic Factors Post-COVID

Team 002 | CSE 6242 Data and Visual Analytics | Spring 2026

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1. Introduction and Problem Statement

We aim to identify which NYC neighborhoods recovered fastest from COVID-19's impact on subway ridership and determine what economic factors predict this recovery. Our analysis covers July 2020 through December 2023 across NYC's Neighborhood Tabulation Areas (NTAs). The NYC Metropolitan Transportation Authority (MTA) and urban planners need neighborhood-level granularity rather than city-wide aggregates to drive equitable investment. We will build an interactive tool to allow stakeholders to visualize these diverse recovery patterns.

2. Current Practice and Limitations

Current analysis lacks interactive, neighborhood-level integration. The MTA provides aggregate statistics but no neighborhood-level recovery analysis, while academic studies typically focus on national trends (Liu et al., 2020) or specific sectors like ride-sharing (Mahfouz et al., 2024). These approaches overlook the specific linkage between subway recovery and neighborhood economics, leaving a gap for a tool that integrates transit recovery data with local economic indicators in an interactive dashboard.

3. Approach and Innovation

We hypothesize that "bouncing back" and "true recovery" are distinct phenomena. We will calculate two metrics for every neighborhood: a **Recovery Index** (growth from 2020 lows) and a **True Recovery Index** (return to pre-pandemic baselines). Our analysis will use k-means clustering to group neighborhoods by trajectory and predictive regression to identify if commercial density or remote work potential drives these patterns. The final deliverable will be an interactive dashboard featuring side-by-side choropleth maps, animated timelines, and drill-down tooltips.

4. Literature Survey

Our methodology adapts three streams of existing research. We use Liu et al. (2020) to establish severity baselines and adapt the place-based methodology of Srinivasan et al. (2025), who linked essential worker concentration to ridership retention. We also build on Che et al. (2023) regarding the resilience of retail clusters and Osorio et al. (2022), whose findings that remote work accounted for 66% of Chicago's rail ridership loss inform our "Remote Work Potential" scoring.

5. Data and Processing

- **MTA Subway Ridership (data.ny.gov):** Hourly station data used for recovery metrics and volume.
- **PLUTO Property Data (NYC Planning):** Tax lot data used to calculate Commercial/Residential land use ratios.

- **LEHD Employment (Census):** Block-level data used for Remote Work Potential scoring.
- **NTA Boundaries (NYC Open Data):** The spatial aggregation unit (262 NTAs).
- **Processing:** We will aggregate ridership to monthly totals, spatially join stations to NTAs, process PLUTO records for land use ratios, and map employment data using a tract-to-NTA crosswalk.

6. Methodology

We will perform coordinate-based spatial joins of subway stations to NTAs, analyzing only those with active service. For economic analysis, NAICS industry codes will be weighted by remote potential (e.g., High for Info/Finance, Low for Retail/Health). We will validate our statistical models using Variance Inflation Factor (VIF) to check for multicollinearity, employ an 80/20 train-test split with 5-fold cross-validation, and use Moran's I to test for spatial autocorrelation.

7. Risks and Mitigations

- **Weak Predictive Signal:** We will test multiple model specifications; results remain valuable for planning even with modest R^2 .
- **Data Quality Issues:** We will cross-validate with multiple sources and strictly document all limitations.
- **Visualization Complexity:** We will use established libraries (Plotly, Folium) and rely on iterative development to ensure usability.

8. Cost

Total Cost: \$0. All data is public; computing will utilize personal machines and free cloud tiers. Software stack includes Python, Plotly, and Folium.

9. Timeline

- **Weeks 1-2 (Isaac, Elias):** Data acquisition, cleaning, and NTA integration.
- **Weeks 3-4 (All):** Recovery index calculation and Exploratory Data Analysis (EDA).
- **Weeks 5-6 (Dami):** Clustering and regression analysis.
- **Weeks 6-7 (David):** Interactive visualization development.
- **Weeks 8-9 (All):** Integration, testing, and iteration.
- **Weeks 10-11 (All):** Final report and poster preparation.

10. Success Metrics

By midterm, we expect all data integrated, recovery indices calculated, and wireframes ready. Our final success criteria include a regression model identifying at least three significant predictors ($p < 0.05$), clustering that achieves meaningful separation of recovery patterns, and a fully functional interactive dashboard that supports seamless map exploration and provides actionable insights for transit planning.

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